Random Variables

Expected Value and Variance

Discrete Distributions

Bernoulli

Geometric

Binomial

Poisson

Continuous Distributions

Uniform

Normal

Student

Exponential

Probability Distributions

Jerome Dumortier

Lecture Overview

Probability distributions as models about uncertain outcomes such as in the case of a city operating a service center:

- How many calls should we expect next hour?
- What is the probability of more than 15 calls?
- How much capacity is needed to handle demand 95% of the time?

Random variables

- Expected value (mean) and variance

Discrete distributions

- Bernoulli, Geometric, Binomial, and Poisson

Continuous distributions

- Uniform, Normal, t or Student, and Exponential

Overview

A random variable is a variable whose value depends on chance

- Number of heads from flipping a coin 20 times
- Number after rolling a die
- Number of passengers showing up to a flight
- Waiting time of a client

Discrete random variables

- A random variable X is discrete if it can assume only a finite or countable infinite number of distinct values

Continuous random variables

- Can take an infinite number of values

Discrete versus continuous random variables

Discrete random variables

- Number of students in a class
- Number of children in a family
- Number of calls to a 911 dispatcher within a 24 hour period

Continuous random variables

- Temperature in a week from today
- Value of the S&P 500
- Average height of IUI students

It is sometimes easier to assume continuity even if the variable seems discrete, e.g., home values in Indianapolis.

Data verssu Probability Distributions

Suppose we observed hourly calls:

```
calls <- c(8, 12, 9, 11, 15, 10, 7, 13, 9, 14)
mean(calls)
```

```
## [1] 10.8
```

Those numbers are data. Representing future calls using a Poisson distribution with a mean of 10.8 represents a model.

Examples of Random Variables

Simulate rolling a die 600 times and the random variable being the number of sixes.

```
dierolls = sample(1:6,size=600,replace=TRUE)
table(dierolls)
```

```
## dierolls
##   1   2   3   4   5   6
##  92 108 112  95  88 105
```

Random Variables and Probability Distribution

A probability distribution is a combination of outcomes of a random variable and associated probabilities. For example, let the random variable X be the number of heads from flipping a coin seven times:

X	0	1	2	3	4	5	6	7
$Pr(X)$	0.01	0.05	0.16	0.27	0.27	0.16	0.05	0.01

The sum of all the probabilities associated with the mutually exclusive outcomes is equal to 1.

Definition

Think of the expected value as a weighted average. If X is a discrete random variable then the expected value of X, i.e., $E(X)$, is written as

$$E(X) = \sum_i x_i \cdot Pr(X = x_i)$$

If X is a continuous random variable, then calculus is needed to calculate the expected value and those details are in the lecture notes. The variance can be calculated as follows:

$$Var(X) = E[(X - E(X))^2] = E(X^2) - E(X)^2$$

Both equations give you the variance. Sometimes one of the equations is more convenient to use. Note that $E(X^2) \neq E(X)^2$.

Example Setup

Suppose you are working for a car dealership. For the last year, you calculated the number of cars sold per day and came up with the following probability distribution:

X	0	1	2	3	4	5
$Pr(X)$	0.10	0.15	0.15	0.30	0.25	0.05

Example Calculations

x_i	$Pr(x_i)$	$x_i \cdot Pr(x_i)$	$x_i - \mu$	$(x_i - \mu)^2$	$Pr(x_i) \cdot (x_i - \mu)^2$
0	0.10	0.00	-2.60	6.76	0.68
1	0.15	0.15	-1.60	2.56	0.38
2	0.15	0.30	-0.60	0.36	0.05
3	0.30	0.90	0.40	0.16	0.05
4	0.25	1.00	1.40	1.96	0.49
5	0.05	0.25	2.40	5.76	0.29
Sum		2.60			1.94

Hence $Var(X) = 1.94$ and $\sigma = \sqrt{1.94} = 1.393$.

Bernoulli Distribution

Characteristics of the Bernoulli distribution:

- Simplest discrete probability distribution
- Two outcomes: "Success" and "Failure"
- One parameter: p

Probability mass function:

$$Pr(X = 1) = p$$

And thus we also have $Pr(X = 0) = 1 - p$.

Random Variables

Expected Value and Variance

Discrete Distributions

Bernoulli

Geometric

Binomial

Poisson

Continuous Distributions

Uniform

Normal

Student

Exponential

Geometric Distribution: Definition

The geometric distribution models the number of trials until the first success in a series of Bernoulli trials.

- How many trials are needed to get the first success?
- Each trial has two outcomes: "Success" (with probability p) or "Failure" (with probability $1 - p$).
- The trials are independent and thus, probability of success remains constant across trials.

Mathematical equation:

$$P(X = k) = (1 - p)^{k-1} \cdot p$$

where X is the number of trials until the first success, p is the probability of success on each trial, and k is the trial number where the first success occurs.

Geometric Distribution: Example

Assume $p = 0.2$, the we have the following:

$$P(X = 3) = (1 - 0.2)^{3-1} \cdot 0.2 = (0.8)^2 \cdot 0.2 = 0.128$$

There is a 12.8% chance the first success occurs on the third trial. Expected value (i.e., average number of trials needed to get the first success) and variance:

$$E(X) = \frac{1}{p}$$

$$Var(X) = \frac{1 - p}{p^2}$$

Geometric Distribution: R

If $p = 0.2$, then we have the following:

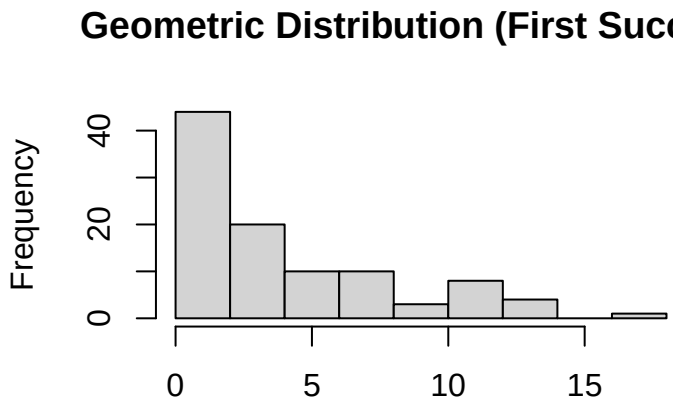
$$E(X) = \frac{1}{0.2} = 5$$

On average, 5 trials for the first success.

```
geometrictrials = rgeom(100,prob=0.2)
```

Geometric Distribution: Histogram

```
hist(geometrictrials,main="Geometric Distribution (First Success)",
     xlab="Number of Attempts",breaks=10)
```



Binomial Distribution: Definition

Characteristics of the Binomial distribution:

- Closely related to the Bernoulli Distribution
- "Repeated" Bernoulli outcomes
- Two parameters: n and p
- k number of success

Probability mass function:

$$Pr(X = k) = \binom{n}{k} \cdot p^k \cdot (1 - p)^{n-k}$$

The mean is $\mu = n \cdot p$.

Binomial: Use

When is the Binomial Distribution appropriate? A situation must meet the following conditions for a random variable X to have a binomial distribution:

- You have a fixed number of trials involving a random process; let n be the number of trials.
- You can classify the outcome of each trial into one of two groups: success or failure.
- The probability of success is the same for each trial. Let p be the probability of success, which means $1 - p$ is the probability of failure.
- The trials are independent, meaning the outcome of one trial does not influence the outcome of any other trial.

Binomial: Example I

Suppose you didn't study for a multiple choice exam. There are 10 questions with five possible answers each. Only one answer per question is correct. What is the probability that you get 6 correct answers?

$$Pr(X = k) = \frac{10!}{6! \cdot (10-6)!} \cdot 0.2^6 \cdot (1-0.2)^{10-6}$$

Or simply in R:

```
dbinom(6,10,0.2)
```

```
## [1] 0.005505024
```

Binomial in R: Probability Density Function

The probability density function (PDF) for the binomial distribution in R is written as `dbinom(x,n,p)`. Consider the following probabilities:

- Probability of 9 heads ($x = 9$) from 16 coin flips ($n = 16$)
- Probability of 0 to 16 heads from 16 coin flips

```
dbinom(9,16,0.5)
dbinom(0:16,16,0.5)
```

Binomial in R: Cumulative density function

The cumulative density function (CDF) for the binomial distribution in R is written as `pbinom(x,n,p)`. Consider the following probabilities:

- Probability of getting up to three heads from flipping a coin ten times
- Cumulative probabilities for getting 0 through 10 heads

```
pbinom(3,10,0.5)
pbinom(0:10,10,0.5)
```

Binomial: Example II

Suppose that 85% of Hoosiers are wearing a seat belt. You are a police officer and pulling over 20 cars. What is the probability that at least (!) 15 people are wearing a seat belt?

```
1-pbinom(14,20,0.85)
```

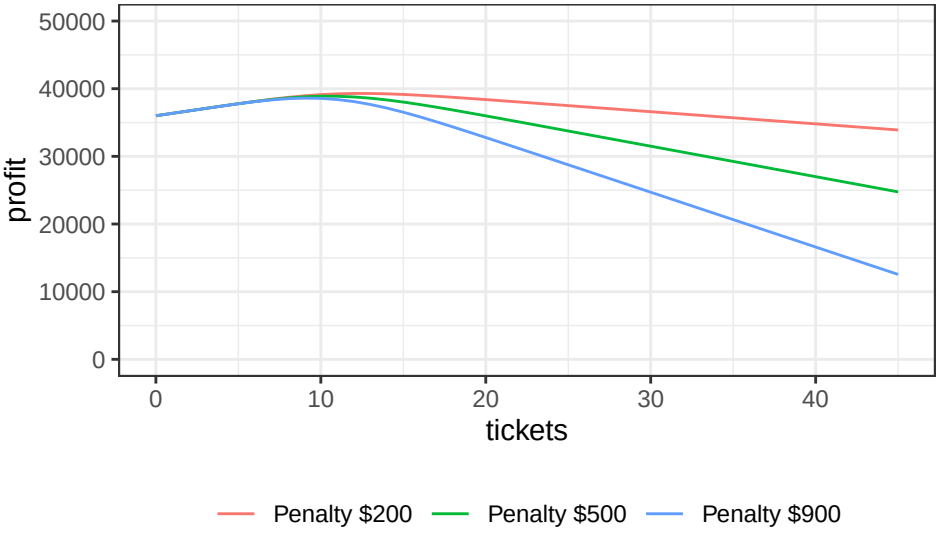
```
## [1] 0.932692
```

While using the binomial distribution, be very careful on how to interpret the results. The probability of at least 15 people wearing a seatbelt means that you are interested in the cumulative probability of 15, 16, 17, 18, 19, and 20 people wearing a seat belt. That probability is 0.933.

Binomial: Overbooking Flights I

The binomial distribution can be used to analyze the issue of overbooking. Assume that an airline as a plane with a seating capacity of 100 The ticket price for each traveler is $400. The airline can overbook the flight, i.e., selling more than 100 tickets, but has to pay $700 in case a person has a valid ticket but needs to be re-booked to another flight. There is a probability of 10% that a booked passenger does not show up. The results for overbooking between 0 and 45 seats are shown on the next slide.

Binomial: Overbooking Flights II



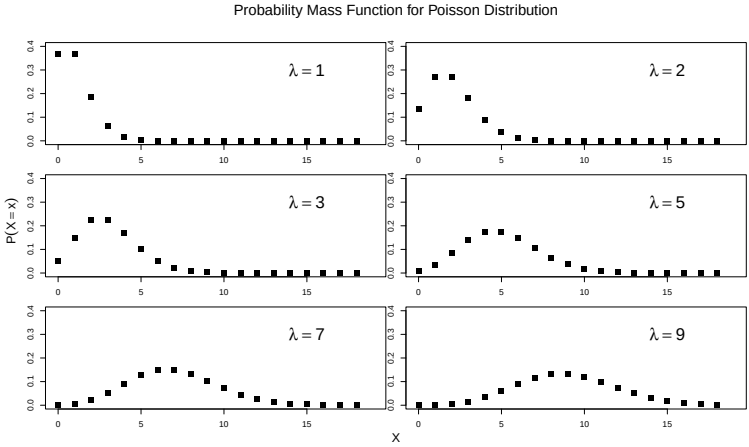
Poisson Distribution

By construction, the Poisson distribution (named after Simeon Denis Poisson, 1781-1840) is used for count data, i.e., $0, 1, 2, \ldots$. The probability mass function for the Poisson distribution is given by:

$$P(X = k) = \frac{\lambda^k \cdot e^{-\lambda}}{k!}$$

An example of the Poisson distribution for different parameter values is shown on the next slide.

Poisson Distribution Example



Poisson Distribution: PDF and CDF

The PDF and CDF of the Poisson Distribution in R are written as `dpois(x,lambda)` and `ppois(x,lambda)`, respectively. Consider the following probabilities:

- Probability of exactly four (x = 4) customers coming to your store when the average is six (lambda = 6)
- Probability of four or less (x = 4) customers coming to your store when the average is six (lambda = 6):

```
dpois(4,6)
```

```
## [1] 0.1338526
```

```
ppois(4,6)
```

```
## [1] 0.2850565
```

Continuous Probability Distributions

Properties:

- Probability of a particular event is zero!
- The area under the probability curve is 1.

Examples

- Uniform distribution
- Bell curve a.k.a. Normal distribution a.k.a. Gaussian Distribution
- Student's t-distribution

Uniform Distribution

The uniform distribution has two parameters, i.e., a and b. If $a <$b, a random variable X is said to have a uniform probability distribution on the interval (a, b) if and only if the density function of X is

$$f(x) = \frac{1}{b-a}$$

Examples:

- $a = 10$ and $b = 40$ then $Pr(25 < x < 30) = 1/6$
- Arrival of your online delivery during your lunch break

Normal Distribution: Introduction

The random variable X is said to be normally distributed with mean μ and variance σ^2 (abbreviated by x $\sim$ N[μ, σ^2] if the density function of x is given by

$$f(x;\mu,\sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \cdot e^{\frac{-1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$

The normal probability density function is bell-shaped and symmetric. The curve is derived from the binomial distribution:

- Galton Board

Standardizing a normal distribution to make it N(0,1) by calculating z, i.e.,

$$z = \frac{X-\mu}{\sigma}$$

z represents the distance from the mean expressed in units of the standard deviation.

Normal Distribution: Example

Suppose that we have a random variable with $\mu = 75$ and $\sigma = 10$. If we are interested in the probability $Pr(60 < x < 70)$ then we have to proceed in three steps:

1. Calculate the probability that $Pr(x < 60)$
2. Calculate the probability that $Pr(x < 70)$
3. Take the difference between the two probabilities

This can be achieved in one step with R:

```
pnorm(70,75,10)-pnorm(60,75,10)
```

```
## [1] 0.2417303
```

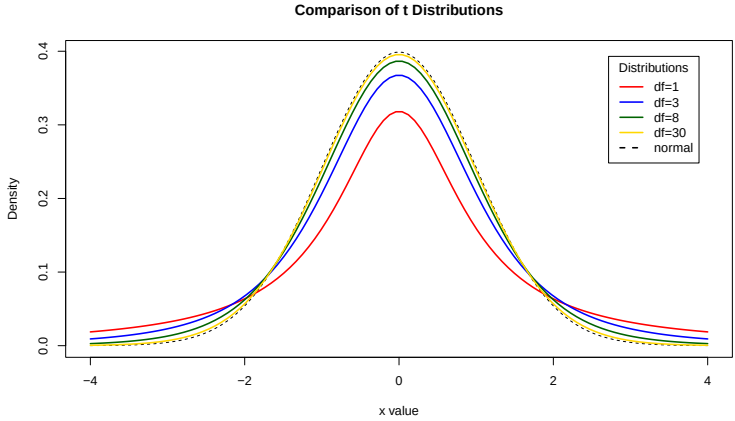
Student Distribution orn t-Distribution: Characteristics

The t-distribution is very similar to the Standard Normal:

- The t-distribution is continuous, symmetric, and bell-shaped.
- The shape (flatness/steepness) depends on the degrees of freedom.
- For very large degrees of freedom (i.e., ∞), the t-distribution is identical to the Standard Normal.

The important aspect of the t-distribution are the tails which are weighted heavier.

Student Distribution orn t-Distribution: Graphical Representation



Exponential Distribution: Overview

If events follow a constant-rate Poisson process, waiting time until the next event can be modeled as

$$T \sim \text{Exponential}(\lambda)$$

with $E(T) = 1/\lambda$. Suppose customers arrive at an average rate of 0.5 per minute:

```
1 / 0.5                          # average wait
1 - pexp(3, rate = 0.5)          # wait > 3 minutes
pexp(4, 0.5) - pexp(1, 0.5)     # 1 < wait < 4
```

Difference Poisson versus Exponential:

- Poisson: How many events?
- Exponential: How long until the next event?

Exponential Assumption: Memorylessness

The Exponential distribution is memoryless:

Under the model, how much longer you expect to wait does not depend on how long you have already waited.

That may be plausible for some random-arrival processes. It is often implausible for:

- Scheduled buses
- Queues that visibly move
- Systems where the event rate changes over time